

No pseudo-Anosov stretch factor is rational

The Last Math Competition, Conjecture 00000000088

Feiyu*

Claude (Opus 5)[†]

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Abstract

Conjecture 00000000088 asserts that every prime $p \geq 2$ is the stretch factor of a pseudo-Anosov element of some mapping class group. It is false — and false for every prime at once, because of a statement that is stronger and cleaner than the negation: *no pseudo-Anosov stretch factor is rational*. A stretch factor λ is an eigenvalue of the action of a homeomorphism on the first cohomology of the double cover that orients the invariant foliations; that action is an automorphism of a lattice, so its characteristic polynomial is monic with constant term ± 1 and λ is an algebraic unit. A rational algebraic unit is ± 1 , whereas $\lambda > 1$. We give the argument in full, and we explain why the Perron–Frobenius candidate matrix recorded in the research notes for this conjecture cannot be repaired: the obstruction is not the determinant of a transition matrix, and no choice of matrix, train track or surface avoids it. The arithmetic half of the proof is formalized in Lean 4 against Mathlib; the geometric half, which needs measured foliations, is classical and is given here by hand.

1 The conjecture as stated

The text of conjecture 00000000088 reads:

English. Conjecture: For every prime $p \geq 2$ there exists a pseudo-Anosov element in some mapping class group whose stretch factor is exactly the integer p .

(The file `conjectures/00000000088.md` gives the same statement in Chinese; we quote the English text only, so that this document compiles without a CJK font.)

2 Reading of the statement

The statement is unambiguous once three conventions are fixed. We fix them in the way that is most generous to the conjecture: every reading we could give it is refuted below.

2.1 Surfaces and mapping classes

Let S be a surface of finite type: a closed surface with finitely many points removed (punctures) or marked, possibly with finitely many boundary components. $\text{MCG}(S)$ is the group of isotopy classes of homeomorphisms of S preserving the set of punctures and the boundary setwise. “Some mapping class group” means $\text{MCG}(S)$ for some such S ; we allow S to be orientable or non-orientable, closed or not, and we place no bound on its genus or on the number of punctures. Infinite-type surfaces are excluded, as they must be: the Nielsen–Thurston classification, and with it the notion of a stretch factor, is a finite-type statement.

*Independent researcher.

[†]Anthropic.

2.2 Pseudo-Anosov maps and stretch factors

Definition 1. A homeomorphism $\varphi: S \rightarrow S$ is *pseudo-Anosov* if there are two transverse measured foliations $(\mathcal{F}^u, \mu^u)$, $(\mathcal{F}^s, \mu^s)$ on S and a real number $\lambda > 1$ with

$$\varphi \cdot (\mathcal{F}^u, \mu^u) = (\mathcal{F}^u, \lambda \mu^u), \quad \varphi \cdot (\mathcal{F}^s, \mu^s) = (\mathcal{F}^s, \lambda^{-1} \mu^s).$$

The number $\lambda = \lambda(\varphi)$ is the *stretch factor* (also *dilatation*) of φ . A mapping class is pseudo-Anosov if it has a pseudo-Anosov representative; the stretch factor depends only on the class [4, 3].

Singularities are allowed: $\mathcal{F}^u$ and $\mathcal{F}^s$ have finitely many singular points with $k \geq 3$ prongs, and $k \geq 1$ prongs at punctures and boundary components. This is the definition in [4, 3, 9], and the one under which the Nielsen–Thurston classification is stated.

2.3 Which number is “the stretch factor”

Two normalizations appear in the literature: the stretch factor λ of Definition 1, which is e^ℓ for ℓ the translation length of φ on Teichmüller space; and the quasiconformal dilatation $K = \lambda^2$ of the Teichmüller representative. The conjecture is refuted under both, and more generally under any convention that names λ^k for a fixed integer $k \geq 1$: see Remark 10. We work with λ .

3 Result

Theorem 2 (Stretch factors are algebraic units). *Let φ be a pseudo-Anosov homeomorphism of a finite-type surface, orientable or not. Then $\lambda(\varphi)$ is an algebraic unit: it is an algebraic integer, and so is $\lambda(\varphi)^{-1}$.*

This is classical — it is the “unit” half of Fried’s theorem that stretch factors are bi-Perron algebraic units [5], and it is used as standard equipment in the literature (e.g. [6, §5]). We give a complete proof in §4 because it is what the refutation rests on.

Theorem 3 (No rational stretch factors). *No pseudo-Anosov stretch factor is a rational number. In particular no integer $n \geq 2$ is a stretch factor.*

Corollary 4. *Conjecture 00000000088 is false, and it fails for every prime $p \geq 2$: for no prime p does any mapping class group contain a pseudo-Anosov element of stretch factor p .*

So the conjecture is not false by an exceptional prime or in a boundary case: the class of numbers it asks for is disjoint from the class of numbers that can occur. The arithmetic content of Theorems 2 and 3 — an eigenvalue of a unimodular integer matrix that is rational equals ± 1 — is formalized in Lean 4 against Mathlib in `lean/Submission/Basic.lean`; see §6 for exactly what is and is not machine-checked.

4 Proof

4.1 The classical input

We use four standard facts. Throughout, Σ denotes the closed surface underlying S , obtained by collapsing each boundary component to a puncture and then filling in all punctures; a homeomorphism of S induces a homeomorphism of Σ permuting the filled points.

(F1) **(Orienting double cover.)** Let φ be pseudo-Anosov on S with invariant foliations $\mathcal{F}^u, \mathcal{F}^s$. There is a double cover $\pi: \hat{S} \rightarrow S$, branched exactly over the singularities with an odd number of prongs, such that $\pi^* \mathcal{F}^u$ and $\pi^* \mathcal{F}^s$ are orientable, and φ lifts to a

homeomorphism $\widehat{\varphi}$ of $\widehat{S}$. The lift is pseudo-Anosov with invariant foliations $\pi^*\mathcal{F}^u, \pi^*\mathcal{F}^s$ and

$$\lambda(\widehat{\varphi}) = \lambda(\varphi).$$

(The cover is the one associated with the orientation character $w: \pi_1(S \setminus \text{Sing}) \rightarrow \mathbb{Z}/2$ of the tangent line field of $\mathcal{F}^u$. Since φ preserves $\mathcal{F}^u$, it preserves w , hence preserves $\ker w$ — a subgroup of index 2, so normal, so unaffected by the basepoint ambiguity of φ_* — and therefore lifts. The transverse measures pull back, and $\pi \circ \widehat{\varphi} = \varphi \circ \pi$ gives $\widehat{\varphi} \cdot \pi^*\mu^u = \pi^*(\varphi \cdot \mu^u) = \lambda \pi^*\mu^u$.) In the flat-geometry language this is the abelian double cover on which the invariant quadratic differential becomes the square of an abelian differential; see [11, §2], [4, Exp. 5, 9], [8].

- (F2) **(Oriented foliations are closed 1-forms.)** An orientable measured foliation on a closed surface Σ , together with a choice of leafwise orientation, is the kernel foliation of a closed 1-form ω with $|\omega| = \mu$, vanishing exactly at the singularities; it therefore has a well-defined class $[\mu] := [\omega] \in H^1(\Sigma; \mathbb{R})$, and for every homeomorphism ψ of Σ ,

$$[\psi \cdot \mu] = (\psi^{-1})^*[\mu].$$

Note that orientability forces every singularity to have an even number of prongs, so the foliation and the form extend across filled punctures and the passage from S to Σ is harmless. See [11, §2], [4].

- (F3) **(Nondegeneracy.)** If $\mathcal{F}^u, \mathcal{F}^s$ is the transverse pair of a pseudo-Anosov map and both are orientable, then the 1-forms of (F2) satisfy

$$\int_{\Sigma} \omega^u \wedge \omega^s = \text{Area} > 0,$$

because $\omega^u + i\omega^s$ is an abelian differential whose horizontal and vertical foliations are $\mathcal{F}^u$ and $\mathcal{F}^s$ and $\omega^u \wedge \omega^s$ is its area form. Hence $[\mu^u] \smile [\mu^s] \neq 0$ in $H^2(\Sigma; \mathbb{R})$, and in particular

$$[\mu^u] \neq 0.$$

- (F4) **(Homological actions are unimodular.)** For a closed surface Σ , $H^1(\Sigma; \mathbb{Z}) \cong \mathbb{Z}^m$ with $m = 2g$, and a homeomorphism ψ acts on it by a group automorphism. In a basis, ψ^* is a matrix $M \in \text{GL}_m(\mathbb{Z})$, so $\det M = \pm 1$.

4.2 Orientable foliations

Proposition 5. *Let φ be a pseudo-Anosov homeomorphism of a finite-type orientable surface S whose invariant foliations are orientable, and let Σ be the closed surface obtained by filling the punctures. Then there are $M \in \text{GL}_m(\mathbb{Z})$ and $\varepsilon \in \{\pm 1\}$ such that $\varepsilon\lambda$ is an eigenvalue of M , where $\lambda = \lambda(\varphi)$.*

Proof. Fix a leafwise orientation of $\mathcal{F}^u$ and let ω^u be the closed 1-form of (F2), so that $[\mu^u] = [\omega^u] \in H^1(\Sigma; \mathbb{R})$. The homeomorphism φ either preserves or reverses the chosen orientation; write $\varepsilon = +1$ in the first case and $\varepsilon = -1$ in the second. Applying (F2) to $\psi = \varphi$ and using $\varphi \cdot \mu^u = \lambda \mu^u$,

$$(\varphi^{-1})^*[\mu^u] = [\varphi \cdot \mu^u] = \varepsilon \lambda [\mu^u].$$

By (F3), $[\mu^u] \neq 0$, so $\varepsilon\lambda$ is an eigenvalue of the $\mathbb{R}$ -linear map $(\varphi^{-1})^*$ on $H^1(\Sigma; \mathbb{R})$. That map preserves the lattice $H^1(\Sigma; \mathbb{Z})$ and is invertible on it, so by (F4) its matrix M in an integral basis lies in $\text{GL}_m(\mathbb{Z})$, and $H^1(\Sigma; \mathbb{R}) = H^1(\Sigma; \mathbb{Z}) \otimes \mathbb{R}$ identifies the two actions. $\square$

4.3 The general case

Proposition 6. *Let φ be a pseudo-Anosov homeomorphism of any finite-type surface, orientable or not, and $\lambda = \lambda(\varphi)$. Then there are $M \in \mathrm{GL}_m(\mathbb{Z})$ and $\varepsilon \in \{\pm 1\}$ such that $\varepsilon\lambda$ is an eigenvalue of M .*

Proof. First let S be orientable. By (F1) pass to the orienting double cover: we obtain a pseudo-Anosov $\widehat{\varphi}$ on the finite-type orientable surface $\widehat{S}$ with orientable invariant foliations and $\lambda(\widehat{\varphi}) = \lambda(\varphi) = \lambda$. Proposition 5 applied to $\widehat{\varphi}$ gives the claim.

If S is non-orientable, let $\rho: \widetilde{S} \rightarrow S$ be its orientation double cover. It is canonical, so every homeomorphism of S lifts; the lift $\widetilde{\varphi}$ of φ preserves the pulled-back measured foliations and scales their measures by the same factors, so $\widetilde{\varphi}$ is pseudo-Anosov on the orientable surface $\widetilde{S}$ with $\lambda(\widetilde{\varphi}) = \lambda$ (see [8]). The orientable case just proved applies to $\widetilde{\varphi}$. $\square$

Remark 7. The torus needs no special treatment. If one admits the Anosov elements of $\mathrm{MCG}(T^2) \cong \mathrm{SL}_2(\mathbb{Z})$ as “pseudo-Anosov”, their stretch factors are by definition eigenvalues of matrices in $\mathrm{SL}_2(\mathbb{Z})$, so the conclusion of Proposition 6 holds for them verbatim, and everything below applies.

4.4 Eigenvalues of unimodular integer matrices

Lemma 8. *Let $M \in \mathrm{GL}_m(\mathbb{Z})$ and let $\alpha \in \mathbb{C}$ be an eigenvalue of M . Then α is an algebraic unit.*

Proof. Let $\chi(X) = X^m + c_{m-1}X^{m-1} + \cdots + c_1X + c_0 \in \mathbb{Z}[X]$ be the characteristic polynomial of M ; it is monic, and $c_0 = \chi(0) = \det(-M) = (-1)^m \det M = \pm 1$. Thus α is an algebraic integer. Moreover $\chi(\alpha) = 0$ gives, after dividing by $\alpha \neq 0$,

$$\alpha^{-1} = -c_0^{-1}(\alpha^{m-1} + c_{m-1}\alpha^{m-2} + \cdots + c_1) = \mp(\alpha^{m-1} + c_{m-1}\alpha^{m-2} + \cdots + c_1),$$

a $\mathbb{Z}$ -polynomial expression in the algebraic integer α ; hence α^{-1} is an algebraic integer as well. $\square$

Proof of Theorem 2. By Proposition 6, $\varepsilon\lambda$ is an eigenvalue of some $M \in \mathrm{GL}_m(\mathbb{Z})$, so $\varepsilon\lambda$ is an algebraic unit by Lemma 8. Algebraic units are closed under negation, so λ is one. $\square$

4.5 Arithmetic

Lemma 9. *A rational algebraic unit is 1 or -1 .*

Proof. Let $\alpha \in \mathbb{Q}$ be an algebraic unit. An algebraic integer in $\mathbb{Q}$ is in $\mathbb{Z}$ ($\mathbb{Z}$ is integrally closed in $\mathbb{Q}$), so $\alpha \in \mathbb{Z}$ and $\alpha^{-1} \in \mathbb{Z}$. Then $\alpha \cdot \alpha^{-1} = 1$ with both factors integers, so α divides 1 in $\mathbb{Z}$, i.e. $\alpha = \pm 1$. $\square$

Equivalently, and this is the form that is formalized: if $\alpha \in \mathbb{Z}$ is a root of a monic $\chi \in \mathbb{Z}[X]$, then $\alpha \mid \chi(\alpha) - \chi(0)$, so $\alpha \mid \chi(0) = \pm 1$.

Proof of Theorem 3. Suppose $\lambda = \lambda(\varphi) \in \mathbb{Q}$ for some pseudo-Anosov φ . By Theorem 2, λ is an algebraic unit, so $\lambda = \pm 1$ by Lemma 9. But $\lambda > 1$ by Definition 1. This is a contradiction, so no stretch factor is rational; integers $n \geq 2$ are in particular rational. $\square$

Proof of Corollary 4. A prime $p \geq 2$ is an integer ≥ 2 , hence is not a stretch factor by Theorem 3. Since p was an arbitrary prime, the conjecture fails for every one of them. $\square$

Remark 10 (the refutation does not depend on the convention). Suppose “stretch factor” is read as λ^k for some fixed integer $k \geq 1$ — for instance $k = 2$, the quasiconformal dilatation $K = \lambda^2$ of the Teichmüller representative. If λ is an algebraic unit then so is λ^k , and $\lambda^k > 1$; so by Lemma 9 the number λ^k is again never rational. Every reading of the conjecture of this shape is false. Likewise, asking for $\lambda = \sqrt{p}$ (so that $K = p$) fails: the minimal polynomial of $\sqrt{p}$ is $X^2 - p$, whose constant term is not ± 1 , so $\sqrt{p}$ is not a unit.

Remark 11 (what the conjecture asks for, in one line). The stretch factors of pseudo-Anosov maps are contained in the bi-Perron algebraic units [5]. A prime p is bi-Perron — trivially, having no other conjugates — but it is not a unit. That single missing condition is what the conjecture overlooks, and it is fatal.

5 Why the Perron–Frobenius candidate cannot be repaired

The research notes accompanying this conjecture propose, for each prime $p \geq 3$, the positive integer matrix

$$M_p = \begin{pmatrix} 2 & 1 \\ p-2 & p-1 \end{pmatrix}, \quad \chi_{M_p}(X) = X^2 - (p+1)X + p = (X-p)(X-1),$$

which is primitive with Perron–Frobenius eigenvalue exactly p , and observe that the associated two-edge substitution $e_1 \mapsto e_1^2 e_2^{p-2}$, $e_2 \mapsto e_1 e_2^{p-1}$ induces M_p on the first homology of the rank-two graph, where $\det M_p = p \neq \pm 1$ rules out a homotopy equivalence. Both computations are correct; they are re-checked formally in the Lean development (`cand_charpoly`, `cand_det`). Three comments place them in context.

1. The Perron–Frobenius side has no obstruction at all. By Lind’s theorem [7], the spectral radii of primitive non-negative integer matrices are exactly the Perron numbers, and every integer $p \geq 2$ is a Perron number. Matrices with Perron root p therefore exist in every sufficiently large size and in great abundance; M_p is one of infinitely many. A search over transition matrices therefore never fails for arithmetic reasons — and, by Theorem 2, never succeeds for geometric ones.

2. The notes’ determinant obstruction does not generalize by itself. A transition matrix of an invariant train track [2] counts, entry by entry, how many times the image of one branch runs over another — *unsigned* crossing numbers. The homological action counts them with signs. The two matrices agree only when no cancellation occurs, which is what happens for the positive substitution above and is why the argument in the notes is valid there. In general the transition matrix is not the homological matrix, and unimodularity is a constraint on the latter, not the former. So the obstruction found in the notes is genuine for that candidate but is not the general mechanism, and one cannot conclude from it that higher-rank train tracks are barred.

3. The general mechanism is Theorem 2, and it closes the problem. Whatever the transition matrix, a pseudo-Anosov map has, after the orienting double cover, a *homological* action that is unimodular and has $\pm\lambda$ among its eigenvalues. So the conclusion of the notes — “no such realization has been established here” — can be strengthened to: no such realization exists, for any train track of any size on any surface, for any prime p . The correct reading of the notes’ determinant computation is that it is the shadow, in the simplest case, of an obstruction that lives one level up.

6 Formalization

The accompanying Lean 4 project (`lean/`, namespace `Conjecture00000000088`, Mathlib `v4.33.1`) formalizes the arithmetic half of the proof — everything from Proposition 6 onwards.

Measured foliations, train tracks and Teichmüller theory are not in Mathlib, so §4.1–§4.3 (the facts (F1)–(F4) and Propositions 5, 6) are *not* machine-checked; they are classical, and are proved above from standard references. We state this division plainly rather than formalizing a restatement of the conjecture that would hide it.

The interface between the two halves is the predicate

`IsUnimodularEigenvalue lam` := there are m , an integer matrix $A \in \mathbb{Z}^{m \times m}$ with `IsUnit A.det`, and a nonzero real vector v with $Av = \text{lam} \cdot v$,

which is exactly the conclusion of Proposition 6 (up to the sign ε , which is immaterial: M may be replaced by $-M$, still unimodular). What is proved in Lean:

- `int_root_dvd_coeff_zero` — an integer root of an integer polynomial divides its constant coefficient.
- `isUnit_charpoly_coeff_zero` — for a unimodular integer matrix, $\chi(0) = \pm \det$ is a unit (Mathlib’s `Matrix.det_eq_sign_charpoly_coeff`).
- `eq_one_or_neg_one_of_charpoly_eval` — an integer eigenvalue of a unimodular integer matrix is ± 1 ; this is Lemma 8 plus Lemma 9 in the integral case.
- `exists_charpoly_map_eval_eq_zero` — a real eigenvector for `lam` makes the characteristic polynomial vanish at `lam` (Mathlib’s `Matrix.exists_mulVec_eq_zero_iff` and `Matrix.eval_charpoly`).
- `rat_eq_one_or_neg_one` — a *rational* number satisfying `IsUnimodularEigenvalue` is 1 or -1 . The descent from $\mathbb{R}$ to $\mathbb{Q}$ to $\mathbb{Z}$ uses that $\mathbb{Z}$ is integrally closed in $\mathbb{Q}$ (`IsIntegrallyClosed.isIntegral_iff`), the characteristic polynomial being monic.
- `irrational_of_isUnimodularEigenvalue` — hence any such number > 1 is irrational: Theorem 3, modulo the geometry.
- `prime_not_isUnimodularEigenvalue` — hence no prime satisfies the predicate: Corollary 4, modulo the geometry.
- `goldenRatio_isUnimodularEigenvalue` — the golden ratio does satisfy it, via $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. The theorems above are therefore not vacuous: what they exclude is rationality and nothing more.
- `cand_charpoly`, `cand_det`, `cand_not_unimodular` — the two computations of the research notes for M_p , and the conclusion that M_p is not unimodular for $p \geq 2$.
- `repaired_det`, `repaired_charpoly`, `repaired_isUnimodularEigenvalue` — the repaired candidate M'_p of §7.1: it is unimodular, its characteristic polynomial is $X^2 - pX + 1$, and its larger root passes, for every integer $p \geq 2$, the test that p itself fails.

Primality is Mathlib’s `Nat.Prime` and irrationality is Mathlib’s `Irrational`; no standard notion is redefined. The project builds with `lake build on leanprover/lean4:v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms of its own: `#print axioms` reports exactly `propxt`, `Classical.choice` and `Quot.sound` for each theorem above.

7 What is true instead

7.1 The nearest true statement

The obstruction is exactly unimodularity, so the natural repair keeps the integer p but moves it from the eigenvalue to the trace. Consider

$$M'_p = \begin{pmatrix} 1 & 1 \\ p-2 & p-1 \end{pmatrix} \quad (\text{the notes' } M_p \text{ with its top-left entry lowered by 1}),$$

so that $\det M'_p = (p-1) - (p-2) = 1$ and $\text{tr } M'_p = p$, whence $\chi_{M'_p}(X) = X^2 - pX + 1$ and

$$\lambda_p = \frac{p + \sqrt{p^2 - 4}}{2}, \quad \lambda_p + \lambda_p^{-1} = p.$$

For $p \geq 3$ this matrix is hyperbolic, and under the isomorphism $\text{MCG}(S_{1,1}) \cong \text{SL}_2(\mathbb{Z})$ it is a pseudo-Anosov mapping class of the once-punctured torus with stretch factor λ_p [3, §13.1]; equivalently it is an Anosov diffeomorphism of T^2 . Thus:

For every prime $p \geq 3$ there is a pseudo-Anosov mapping class whose stretch factor λ satisfies $\lambda + \lambda^{-1} = p$.

This is true, elementary, and is what the conjecture should have asked. It also shows the price of unimodularity precisely: for $p = 3$ one gets $\lambda = (3 + \sqrt{5})/2 \approx 2.618$ rather than 3. The prime $p = 2$ is genuinely exceptional even for the repaired statement — $\text{tr} = 2$ is parabolic, $\lambda = 1$, the mapping class is a Dehn twist and is reducible, not pseudo-Anosov.

7.2 Where the boundary of the possible lies

Fried [5] proved that stretch factors are bi-Perron algebraic units and conjectured the converse: that every bi-Perron algebraic unit has a power that is a stretch factor. That conjecture remains open, and recent work suggests it is false in a strong quantitative sense — asymptotically almost all bi-Perron units of degree at most $2n$ are not stretch factors on a closed genus- n surface [1]. So the exact set of stretch factors is not known; what is known, and all that is needed here, is that it contains no rational number.

7.3 The contrast that makes the conjecture tempting

Integer growth rates are perfectly common one dimension down. By Lind's theorem [7] every Perron number, p included, is the spectral radius of a primitive non-negative integer matrix; and by Thurston's classification of one-dimensional entropies [10], $\log p$ is the topological entropy of a postcritically finite interval map. What fails for surfaces is not the Perron–Frobenius arithmetic but the extra rigidity a surface imposes: the transverse measure of an invariant foliation is a cohomology class of the surface, and a homeomorphism moves cohomology by a unimodular integer matrix.

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